Ethical Intelligence

Python Full Stack Syllabus

Module	Sub-Module / Topics
An Introduction to Python	• A Brief History of Python • Python Versions • Installing Python • Environment Variables • IDLE, Editing Python Files • Python Basic Syntax, Comments • The format Method • Numeric Data Types • Conversion Functions • Simple Output/Input, The % Method • The print Function • Conditional Statements and Logical Operator • The for Loop • The while Loop • break and continue
Python Collections	• Lists • Tuples • Sets • Dictionaries • Sorting Lists and Dictionaries • Copying Collections
Functions, Modules, Comprehension	• Defining Your Own Functions • Parameters • Function Documentation • Keyword and Optional Parameters • Passing Collections to a Function • Variable Number of Arguments • Scope • Passing Functions to a Function • Lambda, Map, Filter, Reduce • Modules, Standard Modules - sys, math, time, The dir Function • List Comprehensions, • Nested List Comprehensions • Dictionary Comprehensions • Specialized Sorts
Exception Handling, Regular Expression, Expressions, Class Objects	• The Exception Handling • Exception Hierarchy, • Handling Multiple Exceptions, raise & assert • Access Modes, Writing Data to a File, Reading Data From a File, Additional File Methods • Classes in Python • Creating Classes • Class Variables, Inheritance, Polymorphism • Simple Character Matches • Special Characters • Quantifiers • The Dot Character • Greedy Matches • Grouping, Matching at Beginning or End • Match Objects

An Overview of Libraries in python	• Numpy • Pandas • Matplotlib • PySpark • Fast API • Beautiful Soup
Framework	• Django
Introduction to SQL	• What is SQL and MSSQL • DBMS vs. RDBMS • Installation of SQL Sever • SQL Data types • Comments • Constraints
DDL and DML commands and operations	• Create Database • Drop Database • Create Table • Drop Table • Truncate • Alter Table • Select • Insert • Update • Delete
Diving into SQL	• Aggregate Functions • SQL Clauses- o Where o Having o Group by o Order by o Union o Union all • SQL joins • Database Designing • Stored Procedures • Indexing
Introduction to HTML, Basics html tags	• Installing code editor • HTML vs. HTML5 • Boilerplate (head and body tags) • HTML Tags and Attributes • Tags vs Element • Title, Link • Heading, Paragraph, Lists • hr and br tags • comments
Diving into HTML	• Images • Hyperlink • div and span tags • Table • Input tags • Select, Textarea • Button • Forms • Header, Footer, Main
Introduction to CSS	• What is CSS? • Difference between inline, internal and external CSS

	• Style Tag, Link Tag • Types of Selectors • Comments
Diving into CSS	• CSS color • CSS background • CSS height / width • CSS box model • CSS border • CSS fonts • CSS display • CSS position
Introduction to JS	• JavaScript introduction • Internal JS vs. External JS • Difference between let, var and const • JS Operators, Datatypes • Prompt and Alert • Array • Objects
Core concepts of JS	• Conditional Statements • Loops • Functions • Array Methods • Map, Filter, Reduce
JavaScript DOM	• DOM Manipulation • JS Events
Advanced JS	• Promises • Fetch API • Async/Await • ES6